MECHENG 451 Homework Assignment

2D Fiber-Scale Resin Impregnation with Temperature and Cure Coupling

Course	MECHENG 451
Assignment	**2D Fiber-Scale Resin Impregnation and Cure Coupling**
Python package	**fiber_impregnation_homework**
Student entry point	student_template/main.py
Main dependencies	Python (numpy, scipy, matplotlib, pandas, Pillow)

Core idea: Fiber pattern + Volume fraction of fiber + Temperature + degree of cure (t) determine resin flow and impregnation.
Geometry controls where resin can flow; cure controls how long resin can continue to flow.

1. Assignment Purpose

This homework uses a transparent 2D thermochemical coupling model to study resin impregnation at the single-fiber scale. Students will connect microstructure, fiber volume fraction (V_f), temperature-dependent viscosity (μ), degree of cure (DoC) polyurethane (PU), gelation, and impregnation behavior. The goal is physical interpretation and computational literacy, not a research-grade multiphase CFD model.

- Represent individual circular fibers explicitly as impermeable obstacles in a 2D rectangular domain.
- Compare square, hexagonal, and random fiber packing patterns.
- Compare V_f = 0.40, 0.50, and 0.60 under a fixed processing temperature.
- Compare isothermal temperature cases from 25 C to 150 C.
- Analyze how cure and gelation change resin mobility over time.

2. Model Scope

The model solves resin flow only through the matrix space between individual fibers. Fiber cells are excluded from the flow domain.

- The 2D fiber area fraction is used as an educational approximation of fiber volume fraction for aligned continuous fibers.
- All temperature cases use the same physical simulation time so animations can be compared directly.
- The cure kinetics and gel degree of cure come from the supplied PU characterization study.
- The cure-viscosity relation and front-propagation model are simplified educational approximations.

3. Student Coding Tasks

Students should work in the student_template/ folder. The geometry generator, pressure solver, time stepping framework, animation engine, plotting framework, and CSV export are already provided. Students only complete the manufacturing-focused geometry, viscosity, and impregnation functions listed below.

Function	Required physics implementation
calculate_fiber_volume_fraction()	Compute $V_f = \frac{sum(\pi r_i^2)}{A_{domain}}$.
calculate_characteristic_spacing()	Compute average nearest-neighbor surface-to-surface fiber spacing.
calculate_viscosity_temperature_only()	Evaluate the Arrhenius temperature-only viscosity relation.
calculate_impregnation_fraction()	Compute $I(t)$ as filled matrix area divided by total non-fiber matrix area.

Where to Fill in Code

Complete only the geometry, viscosity, and impregnation TODO functions below. The cure kinetics and cure-viscosity coupling are supplied; use them for interpretation, not as coding tasks.

File	Functions to complete	What to leave unchanged
student_template/geometry.py	calculate_fiber_volume_fraction() calculate_characteristic_spacing()	fiber pattern generation and rasterization
student_template/viscosity.py	calculate_viscosity_temperature_only()	plotting and parameter-study calls
student_template/impregnation.py	calculate_impregnation_fraction()	front propagation, coupled time stepping, and animation snapshots

After completing the TODOs, run: python fiber_impregnation_homework\student_template\main.py

4. Parameter Studies

Study	Cases and interpretation goals
Packing pattern	At V_f = 0.50 and T = 50 C, compare square, hexagonal, and random packing. Focus on local pathways, bottlenecks, average velocity, and final impregnation.
Fiber volume fraction	For square packing at T = 50 C, compare V_f = 0.40, 0.50, and 0.60. Students should observe how high Vf narrows channels and may stall impregnation early.

Temperature	For square packing at V_f = 0.50, run 25, 50, 75, 100, 125, and 150 C. Compare initial viscosity, cure rate, gel time, final I, and final gelled fraction.
Pattern-temperature coupling	At V_f = 0.50, run square, hexagonal, and random patterns at 25, 75, and 125 C. Compare how geometry and cure compete.

5. Knowledge Points

Topic	What students should understand
Fiber volume fraction	Use 2D area fraction as V_f approximation and distinguish target V_f from achieved V_f.
Packing geometry	Fiber pattern changes connected channel geometry; the code should not hard-code one pattern as better.
Fiber spacing	Surface spacing equals center distance minus the two radii. Smaller spacing increases flow resistance.
Temperature-viscosity relation	The Arrhenius relation gives lower initial viscosity at higher temperature.
Kamal-Sourour cure kinetics	Cure rate depends on temperature and degree of cure through $[k_1 + k_2\, \alpha^m](1-\alpha)^n$.
Gelation	α_{gel} = 0.53 marks the local gel point, after which local resin motion becomes negligible.
Cure-viscosity coupling	The simplified relation $\mu(T,\alpha) = \mu_T(T)\, exp(C_\alpha \alpha)$ makes viscosity rise as cure progresses.
Darcy-like flow	$u = -\left(\frac{K}{\mu}\right)\frac{dp}{dx}$
Saturation field	S(x,y,t) tracks whether resin has reached each matrix cell; fiber cells are excluded from I(t).
Matrix-averaged DoC	DoC plots average alpha over the full matrix area, with dry cells counted as zero, to avoid artificial jumps.

6. Required Outputs

Students should use generated figures, GIF animations, and results_summary.csv to support their written analysis. A strong answer should interpret the geometry and time evolution, not only quote final numerical values.

Output type	Examples
Geometry figures	fiber_patterns.png and spacing_vs_Vf.png.
Thermal and cure figures	viscosity_vs_temperature.png, cure_rate_vs_time.png, degree_of_cure_vs_time.png.
Impregnation figures	impregnation_vs_time_pattern.png,

	impregnation_vs_time_Vf.png, final_impregnation_vs_temperature.png.
Animations	GIFs for square, hexagonal, and random patterns at Vf = 0.50 and selected temperatures.
CSV metrics	I25, I50, I75, I90, final_I, final_average_DoC, final_matrix_average_DoC, gel_time_s.

7. Questions

1. How does increasing Vf affect average fiber spacing?
2. Why do narrower resin channels make impregnation more difficult?
3. At the same Vf, how do square, hexagonal, and random packing change local resin pathways?
4. How does temperature affect the initial resin viscosity?
5. Why does increasing temperature accelerate polyurethane curing?
6. Why can viscosity initially decrease with temperature but later increase during curing?
7. What does α_{gel} = 0.53 physically represent?
8. Compare the moving tracers before and near gelation. What changes?
9. Does the highest tested temperature produce the best impregnation? Explain.
10. Which factor is strongest in the supplied cases: pattern, V_f, or temperature/cure? Support your answer with results.

8. Submission Requirements

- Completed student_template code with the required physics functions implemented.
- Generated results_summary.csv and the required figures/GIF animations.
- A short written report answering the homework questions with physical reasoning.
- Clear distinction between experimentally supplied cure/gel parameters and simplified educational modeling assumptions.

9. Grading Emphasis

- Correct implementation of the governing equations with consistent units.
- Physically reasonable trends: spacing decreases with V_f, initial viscosity decreases with temperature, and cure rate increases with temperature.
- Interpretation based on geometry, viscosity, degree of cure, gelation, and moving tracer animations.
- Recognition that the best processing condition balances fiber architecture, initial viscosity, cure kinetics, and gelation rather than simply maximizing temperature.